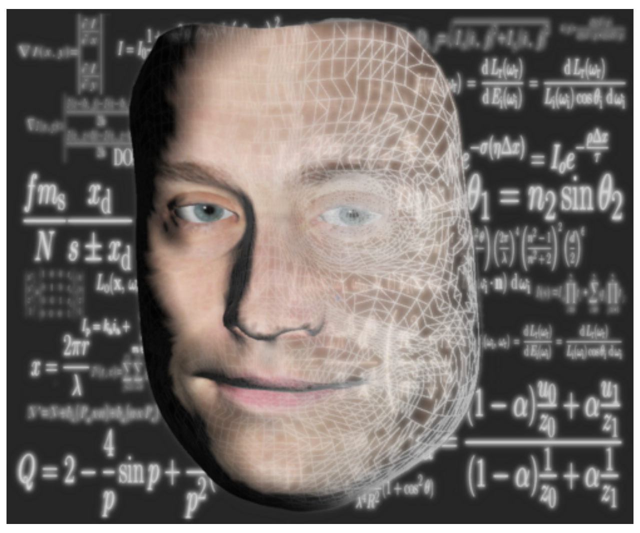

Structural Segmentation with Transformers: SpecTNT for Functional Music Structure Analysis - Extension 2

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1. INTRODUCTION

Music Structure Analysis (MSA) traditionally divides songs into segments with abstract labels (A/B/C) or focuses only on chorus/boundary detection. Explicit semantic labelling (e.g., "verse", "chorus", "bridge") enables powerful applications such as structure-based navigation, automated remixing, live visual synchronization, and real-time analysis of song fragments (e.g., TikTok Clips).

Deep-learning approaches like SpecTNT (Spectral-Temporal Transformer in Transformer) achieve strong results by predicting functional labels directly from audio. However, frame-wise predictions often suffer from fragmentation and temporal inconsistency. This project reimplements the SpecTNT baseline and extends it with a lightweight temporal smoothing decoder to improve section coherence without altering the model architecture.

2. RESEARCH QUESTION

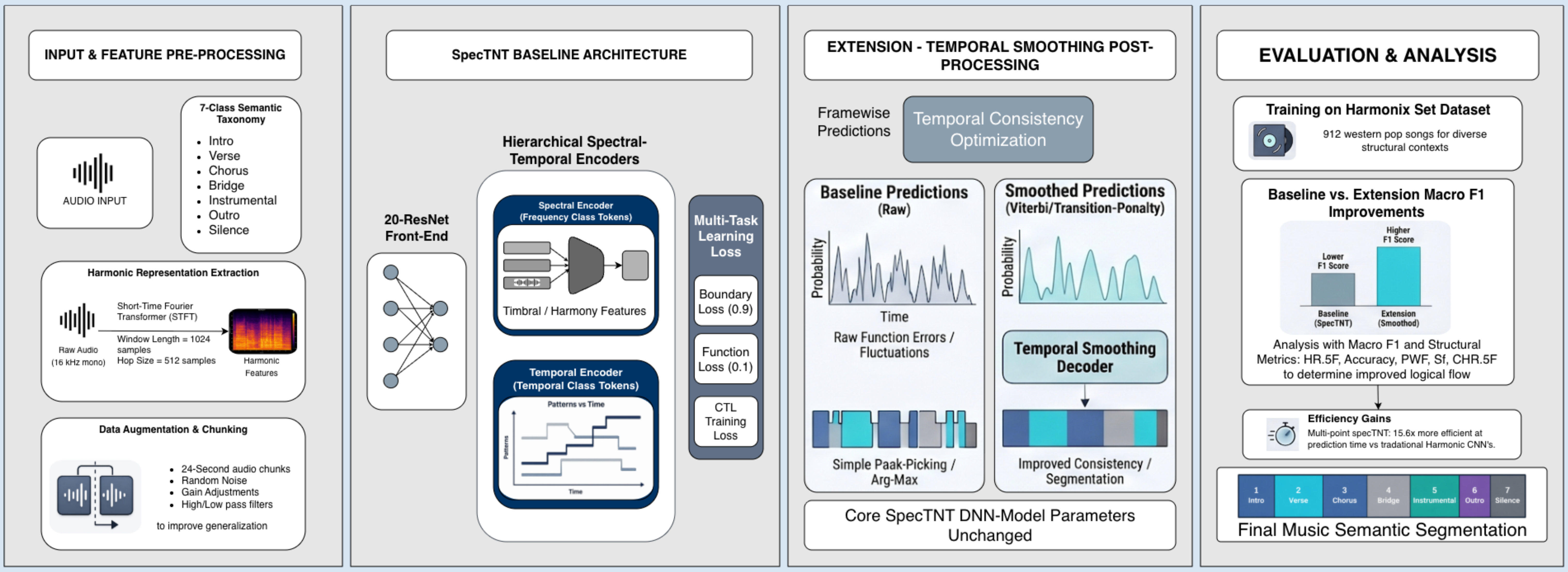
Following Wang et al. (2022), this project performa functional structure analysis using a 7-class taxonomy: *intro, verse, chorus, bridge, instrumental, outro, silence.*

How does applying a temporal smoothing decoder such as Vertigo algorithm or a transition-penalty smoothing on top of SpecTNT's frame wise predictions affect overall accuracy and temporal consistency of predicted musical sections?

Hypothesis: Because the sequence of sections in popular music follows regular patterns, explicitly smoothing frame-wise probabilities during post-processing will significantly improve structural segmentation metrics and macro F1 scores without requiring modifications to the underlying neural network architecture.

3. METHODOLOGY

- **Dataset:** *Harmonix Set.* 912 western pop songs with functional segment annotations.
- **Baseline:** *SpecTNT.*
- **Extension:** *SpecTNT + Temporal Smoothing (Viterbi algorithm or transition-penalty smoothing) post-processing step*
- **Evaluation Metrics:** *Hit Rate (HR.5F), Frame-wise Accuracy (ACC), Pair-Wise F-measure (PWF), Normalized Entropy Score (Sf), Macro F1 - 7 classes, Chorus Hit Rate (CHR.5F)*



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References :

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